Perimeter

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 98 minutes · Total marks: 98

KEY RULES

Perimeter	Total distance around the outside
Rectangle	$P = 2(l + w)$
Circle	$C = \pi d$; semicircle: half the circumference + diameter
Arc	$\frac{\theta}{360} \times \pi d$
Units	Give the unit, e.g. cm

WORKED EXAMPLE 1

[2 marks]

Work out the perimeter of a rectangle 7.5 cm by 4 cm.

Step 1 Perimeter = $2(l + w)$ $2(7.5 + 4) = 23$

Answer: 23 **cm**

WORKED EXAMPLE 2

[3 marks]

Work out the circumference of a circle with radius 5 cm. Give your answer to 1 decimal place.

Step 1 $C = 2\pi r$ $2 \times \pi \times 5 = 31.415\ldots$

Answer: 31.4 **cm**

WORKED EXAMPLE 3

[4 marks]

Work out the perimeter of a semicircle with diameter 12 cm. Give your answer to 3 significant figures.

Step 1 Arc length is half a circle $\frac{1}{2} \times \pi \times 12 = 18.85$

Step 2 Add the straight edge (the diameter) $18.85 + 12 = 30.85$

Answer: 30.8 **cm**

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Perimeter

1 A circle has a diameter of 7.6 cm.
Work out the circumference of the circle.
Give your answer correct to 3 significant figures.

..... cm

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2 A circle has diameter 6.3 cm

Work out the circumference of the circle.
Give your answer correct to one decimal place.

..... cm

(Total for question = 2 marks)

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3 **12** The diagram shows a sector of a circle with radius 7 cm.

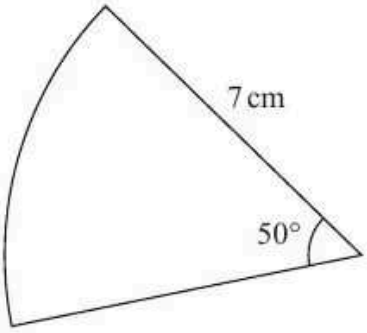


Diagram **NOT** accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

(Total for Question 12 is 2 marks)

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4 The diagram shows a rectangle measuring 10 cm by 3 cm.

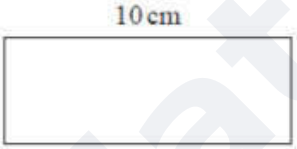


Diagram **NOT** accurately drawn

A shape is made by placing 3 of these rectangles together as shown in the diagram.



Work out the perimeter of the shape.

..... cm

(Total for question = 3 marks)

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5 5 The diagram shows a shaded shape made from three identical semicircles, X, Y and Z

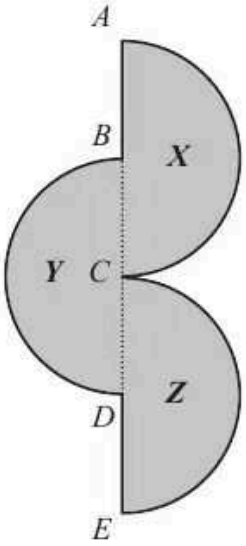


Diagram **NOT** accurately drawn

$ABCDE$ is a straight line.

AC is the diameter of semicircle X and B is the centre of semicircle X
BD is the diameter of semicircle Y and C is the centre of semicircle Y
CE is the diameter of semicircle Z and D is the centre of semicircle Z

$AC = BD = CE = 20$ cm

Work out the perimeter of the shaded shape.
Give your answer correct to the nearest whole number.

..... cm

(Total for Question 5 is 3 marks)

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Perimeter

6 The region, shown shaded in the diagram, is a path.

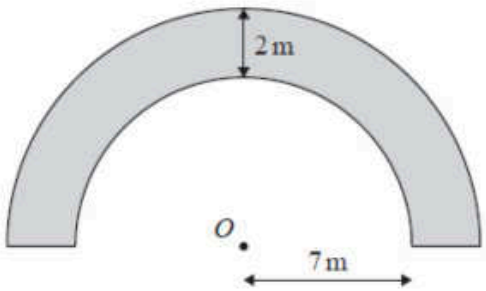


Diagram **NOT** accurately drawn

The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines.

The inner semicircle has a radius of 7 metres.
The path has a width of 2 metres.

Work out the perimeter of the path.
Give your answer correct to one decimal place.

..... m

(Total for question = 3 marks)

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Perimeter

7 OAB is a sector of a circle with centre O and radius r cm.

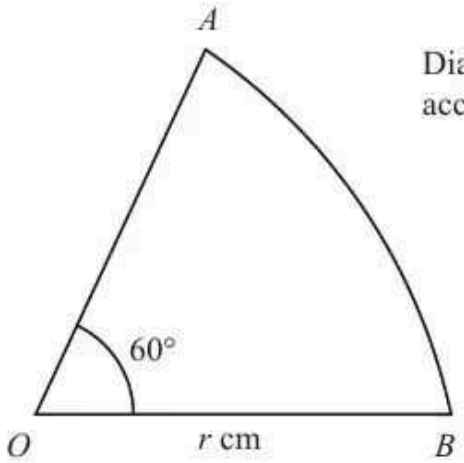


Diagram **NOT** accurately drawn

Angle $AOB = 60°$
The perimeter of the sector is P cm.

Find a formula for P in terms of r
Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

.....

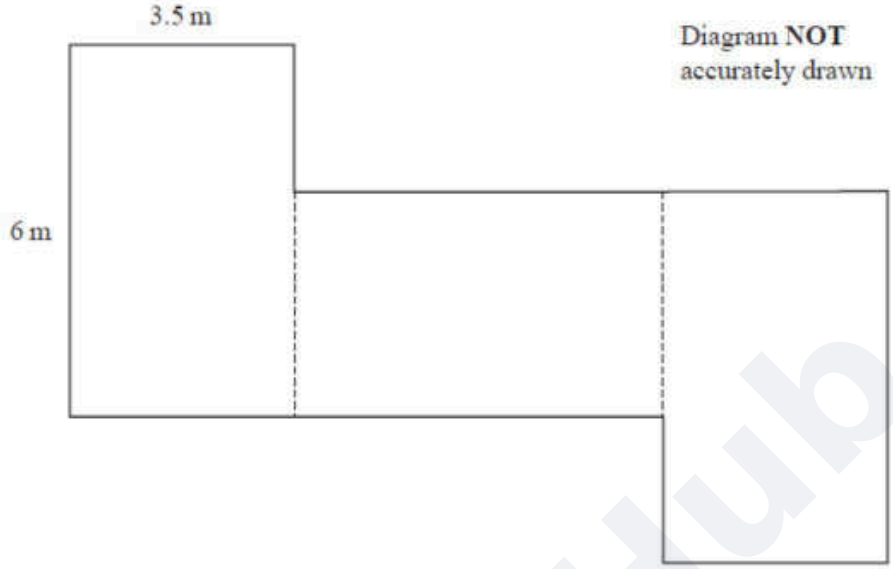
(Total for Question 13 is 3 marks)

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Perimeter

8 The diagram shows a plan of a garden made from three identical rectangles.



The length of each rectangle is 6 metres.
The width of each rectangle is 3.5 metres.

Jeremy puts a fence around the perimeter of the garden.
He charges 7.60 euros for each 1 metre of fence.

Work out how much Jeremy charges in total for the fence.

..... euros

(Total for question = 4 marks)

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Perimeter

9 The diagram shows a shape *ABCDEFG* made from a square *ABDF* and three identical isosceles triangles *BCD*, *DEF* and *FGA*.

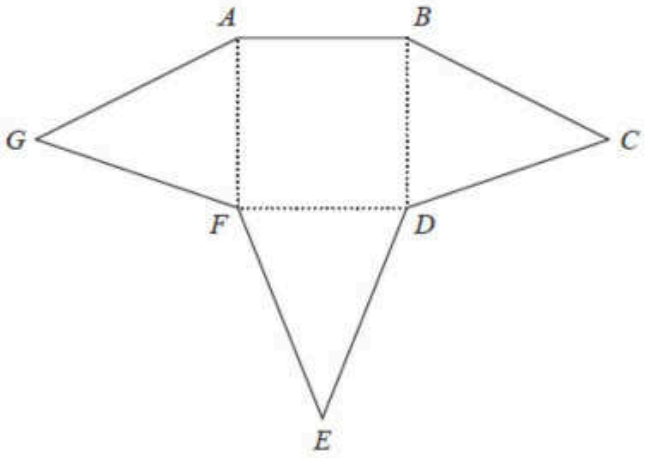


Diagram **NOT** accurately drawn

The perimeter of the square *ABDF* is 48 cm.
The perimeter of each isosceles triangle is 30 cm.

Work out the perimeter of the shape *ABCDEFG*.

..... cm

(Total for question = 4 marks)

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10 4

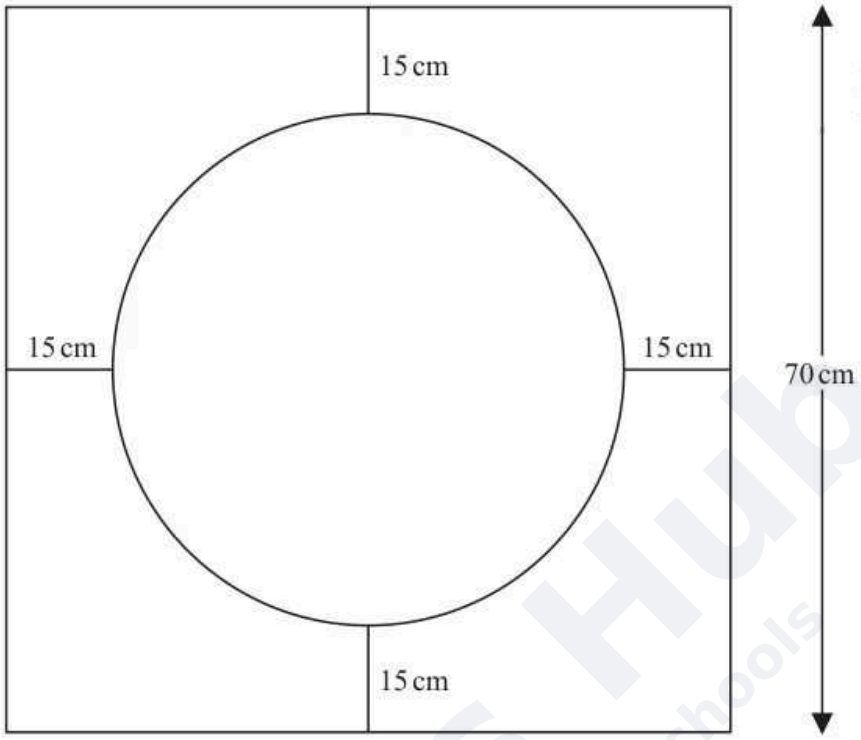


Diagram **NOT** accurately drawn

The diagram shows a design made from wire.

The design is made from
a square with side 70 cm,
a circle with diameter 40 cm,
4 straight pieces each of length 15 cm.

Find the total length of wire needed for the design.
Give your answer correct to the nearest centimetre.

..... cm

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11 **5** The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square $ABCD$

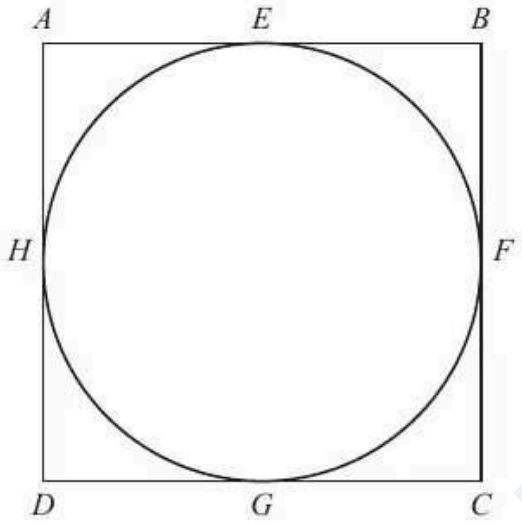


Diagram **NOT** accurately drawn

The circle touches the square at the points E, F, G and H

The area of the square is $81\,\text{cm}^2$

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 5 is 4 marks)

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12 6

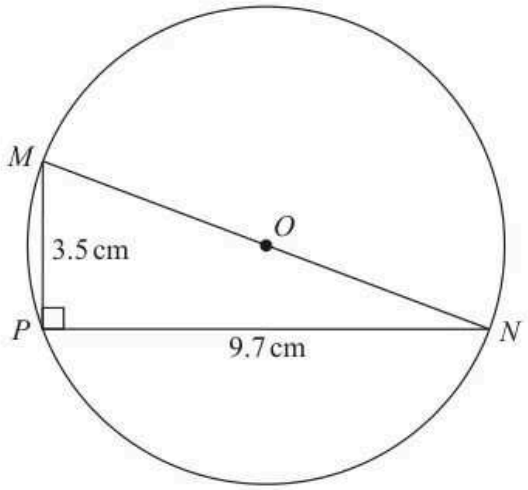


Diagram **NOT** accurately drawn

M, N and P are points on a circle, centre O.
MON is a diameter of the circle.

$MP = 3.5\,\text{cm}$
$PN = 9.7\,\text{cm}$

Angle $MPN = 90°$

Work out the circumference of the circle.
Give your answer correct to 3 significant figures.

..... cm

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13

6 A steam engine for pulling trains has wheels of diameter 1.5 metres.



(a) Calculate the circumference of a wheel.
Give your answer correct to 3 significant figures.

..... m

(2)

The steam engine travels 1000 metres along a test track.

(b) Work out the number of complete turns of a wheel.

.....

(2)

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Perimeter

14 The diagram shows a shaded shape *AEBCD* made by removing triangle *AEB* from rectangle *ABCD*

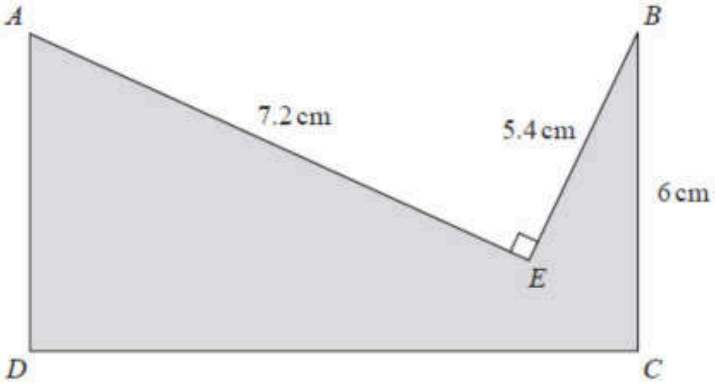


Diagram **NOT** accurately drawn

AE = 7.2 cm *BE* = 5.4 cm *BC* = 6 cm angle *AEB* = 90°

Work out the perimeter of the shaded shape.

..... cm

(Total for question = 4 marks)

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15 **8** The diagram shows an isosceles triangle.

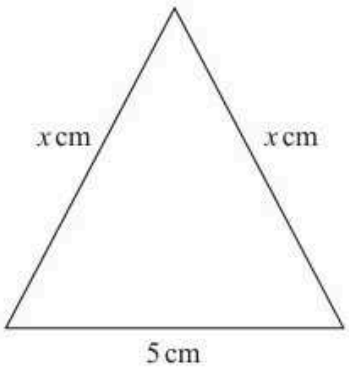


Diagram **NOT** accurately drawn

The area of the triangle is $12\,cm^2$

Work out the perimeter of the triangle.
Give your answer correct to 3 significant figures.

cm

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Perimeter

16 The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

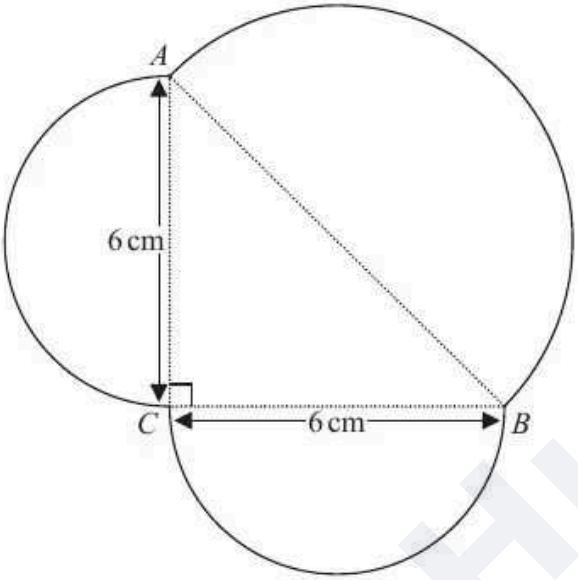


Diagram **NOT** accurately drawn

AB, BC and CA are each the diameter of a semicircle.

$BC = CA = 6$ cm.

Work out the perimeter of the shape.
Give your answer correct to one decimal place.

..... cm

(Total for Question 4 is 5 marks)

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17 The diagram shows a shaded shape $ABCD$ made from a semicircle ABC and a right-angled triangle ACD.

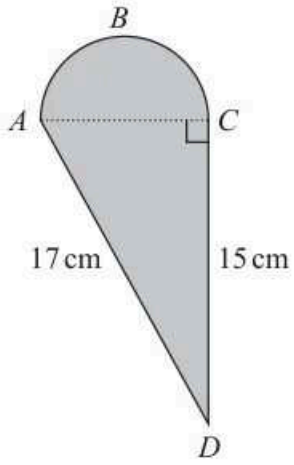


Diagram **NOT** accurately drawn

AC is the diameter of the semicircle ABC.

Work out the perimeter of the shaded shape.
Give your answer correct to 3 significant figures.

..... cm

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18 The diagram shows sector OAB of a circle, centre O.

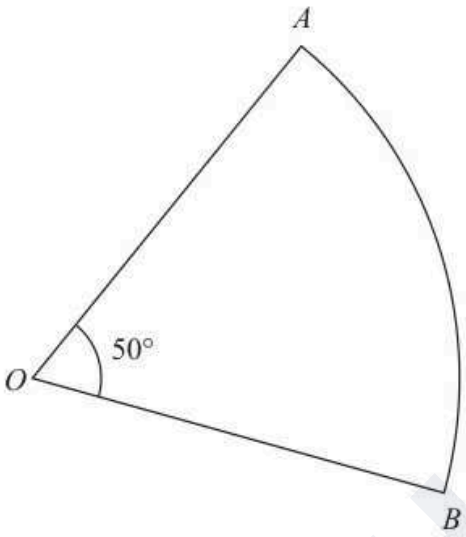


Diagram **NOT** accurately drawn

Angle $AOB = 50°$
Sector OAB has area 20π cm^2

Calculate the perimeter of sector OAB.
Give your answer correct to 3 significant figures.

.....cm

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19 **8** The diagram shows a trapezium, $ABCD$

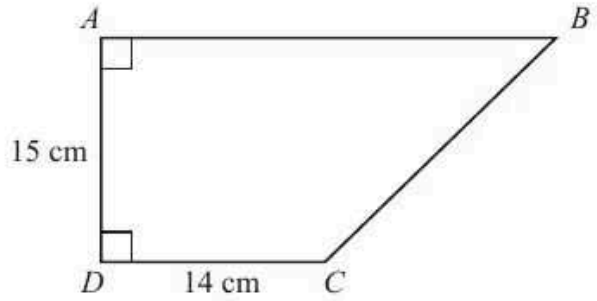


Diagram **NOT** accurately drawn

DAB and ADC are right angles.

$AD = 15$ cm $DC = 14$ cm

The area of the trapezium is 360 cm²

Work out the perimeter of the trapezium.

..... cm

(Total for Question 8 is 6 marks)

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CHALLENGE QUESTIONS

20 The diagram shows a sector $OAPB$ of a circle, centre O.

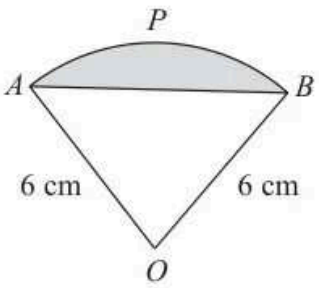


Diagram **NOT** accurately drawn

AB is a chord of the circle.
$OA = OB = 6$ cm.

The area of sector $OAPB$ is 5π cm^2

Calculate the perimeter of the shaded segment.
Give your answer correct to 3 significant figures.

..... cm

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21 A circle centre O has radius 9 cm.

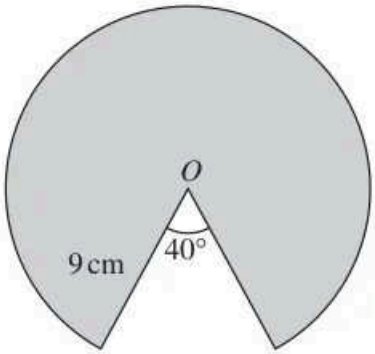


Diagram **NOT** accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

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Perimeter

22 The diagram shows a shaded shape *ABCDEO* made from a semicircle *BCD* and two identical right-angled triangles *ABO* and *EDO*

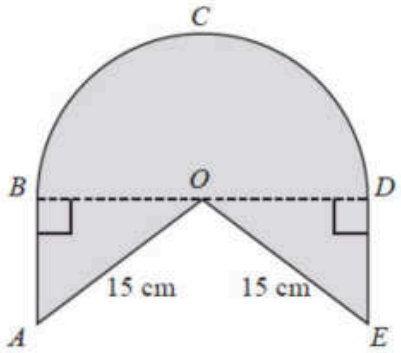


Diagram **NOT** accurately drawn

BD is the diameter of the semicircle.
O is the midpoint of *BD*

angle *ABO* = angle *EDO* = 90° *AO* = *EO* = 15 cm

The area of semicircle *BCD* is 227 cm^2

Work out the perimeter of the shaded shape.
Show your working clearly.
Give your answer correct to one decimal place.
..... cm

(Total for question = 5 marks)

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23 *ABCED* is a five-sided shape.

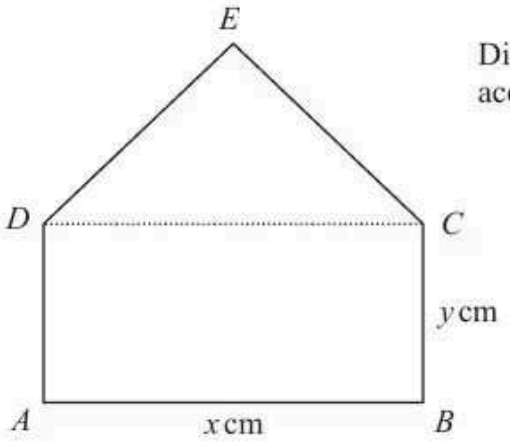


Diagram **NOT** accurately drawn

ABCD is a rectangle.
CED is an equilateral triangle.

$AB = x$ cm $BC = y$ cm

The perimeter of *ABCED* is 100 cm.
The area of *ABCED* is R cm^2

(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ .....

(2)

ii) Explain why the maximum value of R is given by this value of x.

.....

(1)

(Total for Question 19 is 6 marks)

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24

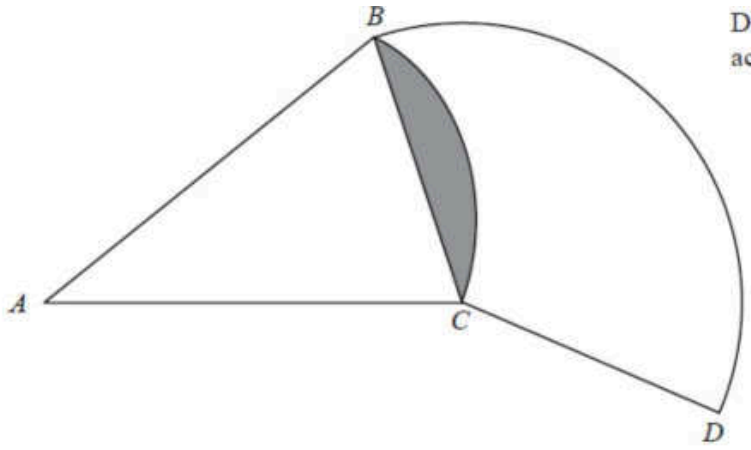


Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle *BAC* = 40°
Angle *BCD* = 130°
Area of shaded segment = 28 cm^2

Find the length of the arc *BD*
Give your answer correct to 3 significant figures

..... cm

(Total for question = 6 marks)

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TOTAL FOR PAPER IS 98 MARKS